You are given an array of integers `heights` where `heights[i]` represents the height of a bar. The width of each bar is `1`.

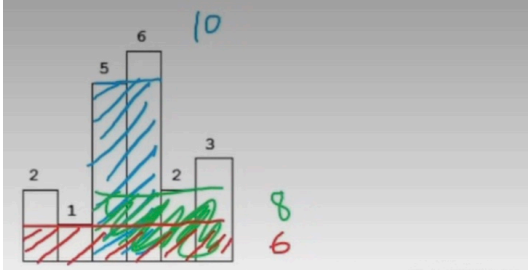
Return the area of the largest rectangle that can be formed among the bars.

Note: This chart is known as a histogram.

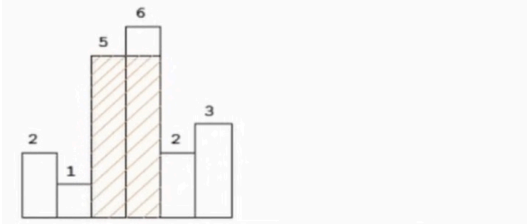
84. Largest Rectangle in Histogram

Hard 3356 78 Add to List Share

Given *n* non-negative integers representing the histogram's bar heigh where the width of each bar is 1, find the area of largest rectangle in t histogram.



Above is a histogram where width of each bar is 1, given height = `[2,1,5,6,2,3]`.

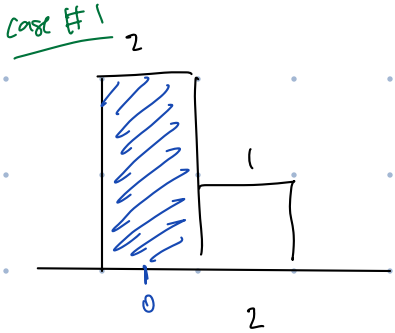


The largest rectangle is shown in the shaded area, which has area = `10` unit.

Example:

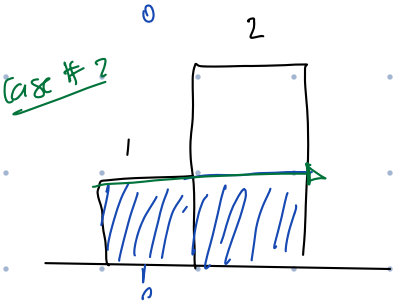
```
Input: [2,1,5,6,2,3]
Output: 10
```

case # 1



From the right side the max height is 2 & width is 1 in order for this to produce max-area.

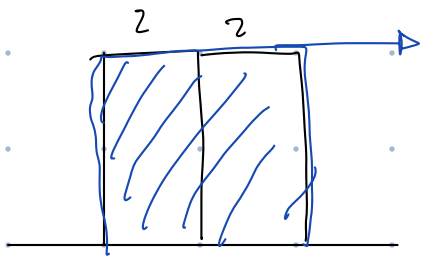
case # 2



From the right max height is 1 & width is 2.

Need to go both direction.

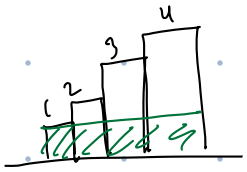
case # 3 : even height.



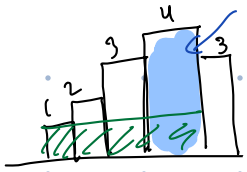
No "hole", so extend further to right.

Q Current heights are increasing order, they will be popped. b/c in case # 1, 2 cannot be extended farther.

Case #4: increasing order.



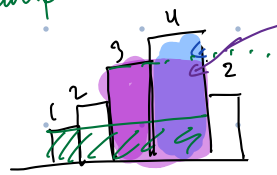
4 can be extended farther, but say you have a smaller right after.



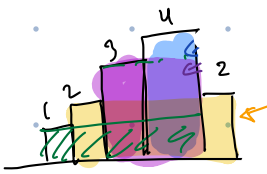
compute area of 4 & then pop it.

remove from consideration.

Examp: what if we have a 2 instead of a 3.



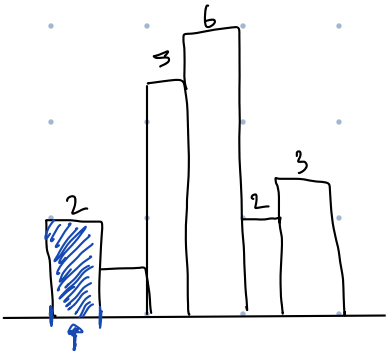
compute area & then pop it.



extending is possible.

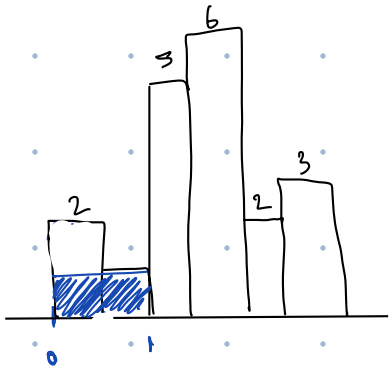
We only pop elements that are most recent. That is from the top; hence, we can use a stack.

Example. — Run through.



stack	
index	height
0	2

max Area.
2

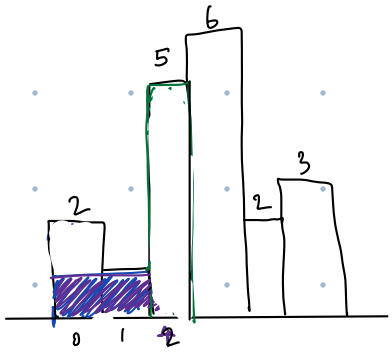


stack	
index	height
~~0~~	~~2~~
0	1

← pop b/c it is greater than current height.

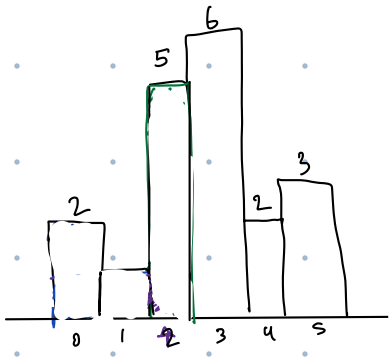
max Area.
2

← update max height is still 2.



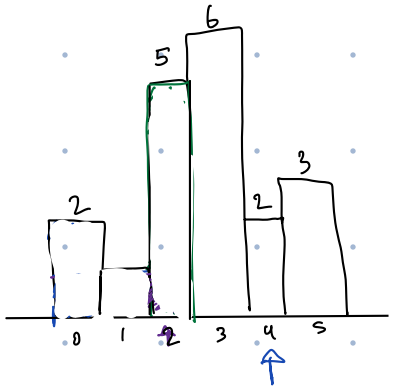
stack	
index	height
~~0~~	~~2~~
0	1
2	5

max Area.
2



stack	
index	height
~~0~~	~~2~~
0	1
2	5
3	6

max Area.
2



stack	
index	height
~~0~~	~~2~~
0	1
2	5
3	6
	2

max Area.
2

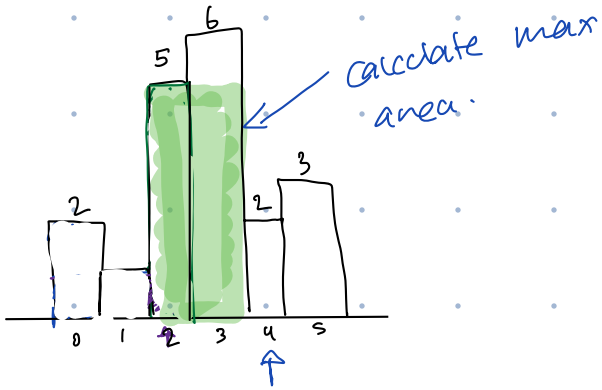
Since 2 is smaller than 6, we have to pop 6.

Before doing so, we need to calculate the area that the area of 6 could have created.

stack	
index	height
~~0~~	~~2~~
0	1
2	5
~~3~~	~~6~~
	2

max Area.
~~2~~
6

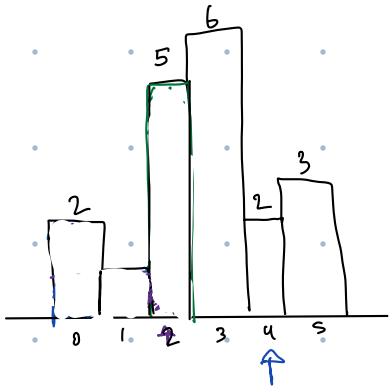
Great, but now $5 > 2$, that means we have to also pop it. It starts at 2 & stops at 4 (and the corner #s are at).



stack	
index	height
~~0~~	~~2~~
0	1
~~2~~	~~5~~
~~3~~	~~6~~
	2

max Area.
~~2~~
~~6~~
10

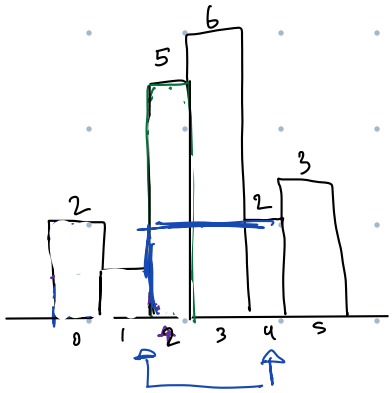
Now look at the top of the stack. It is 1, since that is smaller that 2, no need to pop it. The 1 can continue to be extended.



stack	
index	height
~~0~~	~~2~~
0	1
~~2~~	~~5~~
~~3~~	~~6~~
2	

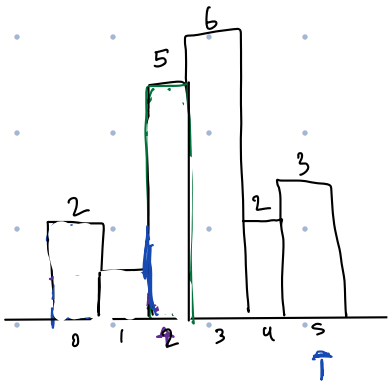
max Area.
~~2~~
~~6~~
10

The index of height 2, can be 4 but it can start at 1 b/c we popped two elements, meaning that it can be extended.



stack	
index	height
~~0~~	~~2~~
0	1
~~2~~	~~5~~
~~3~~	~~6~~
2	2

max Area.
~~2~~
~~6~~
10



stack	
index	height
~~0~~	~~2~~
0	1
~~2~~	~~5~~
~~3~~	~~6~~
2	2 or
5	3

max Area.

~~2~~

~~6~~

10

3 > 2, so we do not need to pop it. B/C we cannot extend backwards anymore, then the start index is 5.

Now, we have 3 elements still in stack, meaning we were able to extend all the way to end of histogram.

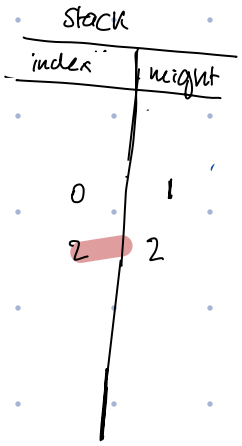
Thus, we need to iterate these positions & calculate areas.

stack	
index	height
0	1
2	2
5	3

at index 5 it started & went all way to end, so lenght is 1.

area = 3.

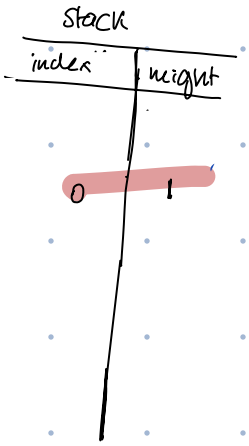
3 < 10, so no update.



at index 2 it started & went all way to end, so lenght is 4

area = 8

8 < 10, so no update.



at index 0 it started & went all way to end, so lenght is 6

area = 6

6 < 10, so no update.

Thus max area is 10.

We iterate elements once. Push elements once & pop once, meaning the overall time complexity is O(n).

Worst case stack space complexity is O(n).